ENGR 484 - Design Project

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ENGR 484 Design Project

gas to air single pass crossflow HX

both flows are single pass and unmixed

$K_f = 18 \, W/m\cdot K$; $\delta_w = 0.5 \, mm$; $K_c = 0.36$, $K_e = 0.42$

$\varepsilon = 0.8381$

$T_{h,i} = 900°C$ (gas)

$T_{c,i} = 200°C$ (air)

$\Big\}$ use air properties for both

$\dot{m}_h = 1.66 \, kg/s$

$\dot{m}_c = 2 \, kg/s$

Pressure drops limited to : $\Delta P_h = 9.05 \, kPa$

$\Delta P_c = 8.79 \, kPa$

$P_{h,i} = 160 \, kPa$

$P_{c,i} = 200 \, kPa$

$b = 2.49 \, mm$, $D_h = 1.54 \, mm$, $\alpha = 941.6 \, \frac{m^2}{m^3}$, $\delta_f = 0.102 \, mm$

$\left(\frac{A_f}{A}\right) = 0.785$

$\rho = 0.782/mm$; $l_f = 3.18 \, mm$; $r_h = \frac{D_h}{4} = 0.385 \, mm$

fin density

fin offset length

- Computing T_m's and determining properties

estimating c_p:

hot gas ($900°C = 1173K$) | cold air ($200°C = 473K$)

$c_{p,h} = 1170.68\ J/kg\cdot K$ | $c_{p,c} = 1025.14\ J/kg\cdot K$

$C_h = \dot{m}_h c_{p,h} = 1943.33\ W/K$ | $C_c = \dot{m}_c c_{p,c} = 2050.28\ W/K$

→ $C_{min} = C_h$ $\quad C_r = 0.95$

now,

$$q = \varepsilon C_{min} (T_{h,i} - T_{c,i})$$

$T_{h,o}$: $\varepsilon C_{min} (T_{h,i} - T_{c,i}) = C_h (T_{h,i} - T_{h,o})$

$T_{h,o} = 313.33°C$

$T_{c,o}$: $\varepsilon C_{min} (T_{h,i} - T_{c,i}) = C_c (T_{c,o} - T_{c,i})$

$T_{c,o} = 756.07°C$

from table: (C_{min} = hot, $C_r = 0.95$)

$$T_{h,m} = \frac{(T_{h,i} + T_{h,o})}{2} = 606.67°C = 879.67K$$

$$T_{c,m} = \frac{(T_{c,i} + T_{c,o})}{2} = 478.04°C = 751.04K$$

$T_{h,m} = 879.67\ K$

$\rightarrow C_{P,h} = 1116.53\ J/kg\cdot K$

$C_h = 1853.44\ W/K$

as $C_h = C_{P,h} \cdot \dot{m}_h$

$C_{min} = C_h$

$T_{c,m} = 751.04\ K$

$\rightarrow C_{P,c} = 1087.25\ J/kg\cdot K$

$C_c = 2174.5\ W/K$

as $C_c = C_{P,c} \cdot \dot{m}_c$

for $T_{h,o}$:

$$\varepsilon C_{min}(T_{h,i} - T_{c,i}) = C_h(T_{h,i} - T_{h,o})$$

$$T_{h,o} = 313.33°C = 586.33\ K$$

for $T_{c,o}$:

$$\varepsilon C_{min}(T_{h,i} - T_{c,i}) = C_c(T_{c,o} - T_{c,i})$$

$$T_{c,o} = 700.05°C = 973.05\ K$$

$$T_{h,m} = \frac{(T_{h,i} + T_{h,o})}{2} = 606.67°C = 879.67\ K$$

$$T_{c,m} = \frac{(T_{c,i} + T_{c,o})}{2} = 450.03°C = 723.03\ K$$

hot gas (879.67 K):

$\rho = 0.3961 \frac{Kg}{m^3}$, $C_p = 1116.5 \frac{J}{KgK}$, $\mu = 3.92 \times 10^{-5} \frac{N \cdot s}{m^2}$, $K = 0.06102 \frac{W}{mK}$

$Pr = 0.718$

cold air (723.03 K):

$\rho = 0.4822 \frac{Kg}{m^3}$, $C_p = 1080.53 \frac{J}{KgK}$, $\mu = 3.46 \times 10^{-5} \frac{Ns}{m}$, $K = 0.05355 \frac{W}{mK}$

$Pr = 0.698$

- Determining NTU and ntu

$C_r = \frac{C_{min}}{C_{max}} = 0.85$, $\varepsilon = 0.8381$

From effectiveness-NTU relations

cross flow (single pass), both fluids unmixed

$$\varepsilon = 1 - \exp\left[\left(\frac{1}{C_r}\right)(NTU)^{0.22}\left\{\exp\left[-C_r(NTU)^{0.78}\right] - 1\right\}\right]$$

solving,

$NTU \approx 7.79$

for gas-gas HX

$ntu_h \approx ntu_c \approx 2NTU$

$ntu_h = 15.58$

$ntu_c = 15.58$

- obtaining approximate mean values of j_H/C_f over the complete Re range

from plot of (j_H/C_f) vs Re, taking 15 data points

$\Rightarrow$ 0.215, 0.23, 0.24, 0.25, 0.258, 0.261, 0.269, 0.273, 0.281, 0.285, 0.288, 0.289, 0.291, 0.294, 0.295

mean = 0.268

$(j_H/C_f) = 0.268$

- Computing mass velocity G's (assume $\eta_o = 0.8$)

$\Delta P_h = 9.05$ KPa, $\Delta P_c = 8.79$ KPa

density at inlets and outlets:

$$\bar{R} = \frac{R}{M} = 287.1015$$

hot gas,

$$\rho_i = \frac{P_i}{\bar{R} T_i} = 0.4751 \text{ Kg/m}^3$$

$$\rho_o = \frac{P_o}{\bar{R} T_o} = 0.9504 \text{ Kg/m}^3$$

cold air,

$$\rho_i = \frac{P_i}{\bar{R} T_i} = 1.4728 \text{ Kg/m}^3$$

$$\rho_o = \frac{P_o}{\bar{R} T_o} = 0.7160 \text{ Kg/m}^3$$

now,

$$G = \left[\frac{2}{(1/\rho)_m \cdot Pr^{2/3}} \cdot \frac{\eta \cdot \Delta P}{ntu} \left(\frac{j_H}{C_f} \right) \right]^{1/2}$$

where

$$\left(\frac{1}{\rho}\right)_m \approx \frac{1}{2}\left(\frac{1}{\rho_i} + \frac{1}{\rho_o}\right)$$

$G_h = 14.028$

$G_c = 17.2118$

- Determining j_H and C_f

$$Re_h = \frac{G_h \cdot D_h}{\mu_h} = 551.1$$

$$Re_c = \frac{G_c \cdot D_h}{\mu_c} = 766.1$$

From fig. 3

for gas flow (hot): $j_H = 0.0175$, $C_f = 0.07$

for cold flow (air): $j_H = 0.015$, $C_f = 0.056$

- Determining overall heat transfer coefficient

$$h = j_H \cdot G \cdot c_p \cdot Pr^{-2/3} \; ; \; \eta_f = \frac{\tanh(ml)}{(ml)} \; , \; m = \left(\frac{hP}{K_f \cdot A_K} \right)^{1/2} , \; \eta_o = 1 - (1 - \eta_f) \frac{A_f}{A}$$

$P = 2(l_f + \delta_f) = 6.564 \, mm$

$A_K = l_f \cdot \delta_f = 0.32436 \, mm^2$

$l = \frac{b}{2} - \delta_f = 1.143 \, mm$

for hot gas,

$h_h = 338.5\ W/m^2 \cdot K$

$m = 616.9\ m^{-1}$

$\eta_f = 0.8167$

$\eta_o = 0.8561$

for cold air,

$h_c = 354.53\ W/m^2 \cdot K$

$m = 631.34\ W/m^2 \cdot K$

$\eta_f = 0.8563$

$\eta_o = 0.8872$

assuming $R_f'' = 0$

$$\frac{1}{U_h} = \frac{1}{(\eta_o h)_h} + \frac{\alpha_h}{\alpha_c}\left(\frac{1}{(\eta_o h)_c}\right)$$

$U_h = 150.83\ W/m^2 \cdot K$

• Calculating core dimensions

$$A_h = \frac{NTU \cdot C_{min}}{U_h} = 95.73\ m^2$$

$$\sigma = \alpha \cdot \frac{D_h}{4}$$
$$= 0.363$$

$$A_c = \frac{\alpha_c}{\alpha_h} \cdot A_h \Rightarrow A_c = 95.73\ m^2$$

$$A_{o,h} = \left(\frac{\dot{m}}{G}\right)_h = 0.1183\ m^2 \quad ; \quad A_{o,c} = \left(\frac{\dot{m}}{G}\right)_c = 0.1162\ m^2$$

$$A_{fr,h} = \frac{A_{o,h}}{\sigma_h} = 0.326\ m^2 \quad ; \quad A_{fr,c} = \frac{A_{o,c}}{\sigma_c} = 0.320\ m^2$$

• Core lengths

$$L_g = \left(\frac{D_h \cdot A}{4 \cdot A_o}\right)_h = 0.311\ m$$

$$L_a = \left(\frac{D_h \cdot A}{4 \cdot A_o}\right)_c = 0.317\ m$$

$$L = \frac{A_{fr,h}}{L_a} = 1.0284\ m$$

• Computing pressure drops

$$\Delta P = \frac{G^2}{2\rho_i}\left[1-\sigma^2+K_c+2\left(\frac{\rho_i}{\rho_o}-1\right)+C_f\cdot\frac{L}{r_h}\cdot\rho_i\left(\frac{1}{\rho}\right)_m-(1-\sigma^2-K_e)\frac{\rho_i}{\rho_o}\right]$$

for hot gas,

$\Delta P_h =$ ~~977~~ 8783.1 Pa

for cold air,

$\Delta P_c = 7331.5$ Pa

needs iteration

→ Reiteration will be done in MATLAB by calculating new values of G's and also considering thermal resistance in wall.

$$G = \left\{2\rho_i\Delta P \Big/ \left[1-\sigma^2+K_c+2\left(\frac{\rho_i}{\rho_o}-1\right)+C_f\frac{L}{r_h}\rho_i\left(\frac{1}{\rho}\right)_m-(1-\sigma^2-K_e)\frac{\rho_i}{\rho_o}\right]\right\}^{1/2}$$

$$\frac{1}{U_h} = \frac{1}{(\eta_o h)_h} + \frac{\delta_w A_h}{k_w A_w} + \frac{1}{(\eta_o h)_c}$$

MATLAB code will be provided in submission to show the iteration

From MATLAB iteration:

$\Delta P_{gas, specified} = 9050$ Pa
$\Delta P_{air, specified} = 8790$ Pa

iteration	ΔP_{gas} (Pa)	ΔP_{air} (Pa)	|error %| ΔP_{gas}	|error %| ΔP_{air}
1	8663.3	8994.7	4.270	2.330
2	9075.4	8534.2	0.280	2.910
3	8939.3	8822.8	1.220	0.373
4	9088.3	8757.8	0.420	0.370

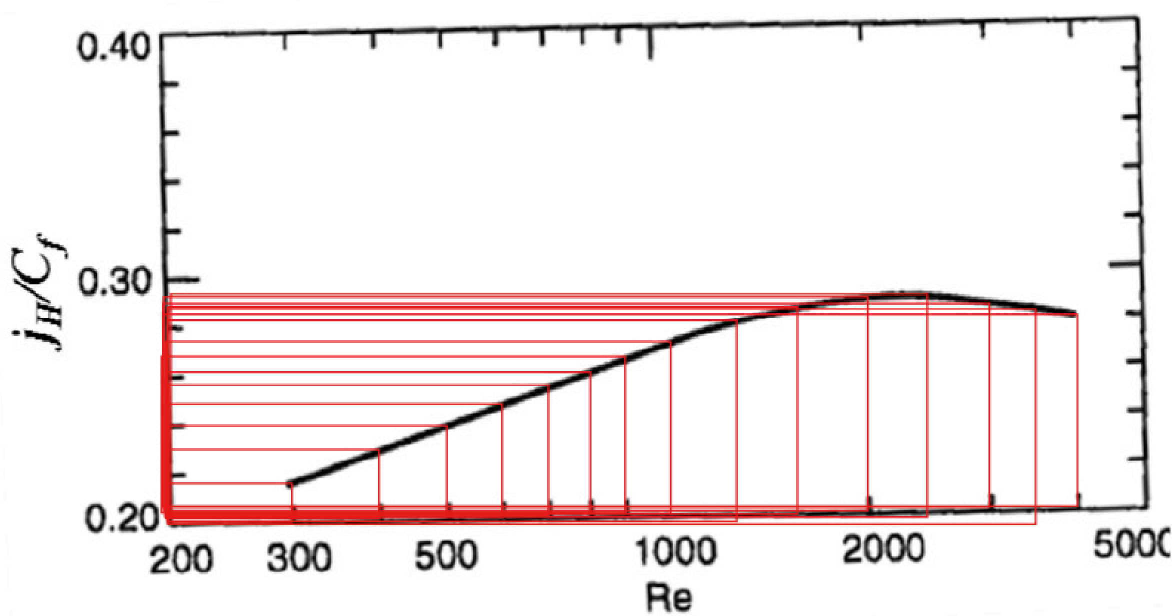
In the 4^{th} iteration, both Pressure drops for air and gas are below 1% so we pick the core dimensions from the 4^{th} iteration

$L_g = 0.3049 m \approx 0.305 m$

$L_a = 0.3266 m \approx \cancel{0.33}\ 0.327 m$

$L = 0.9599 m \approx 0.960 m$

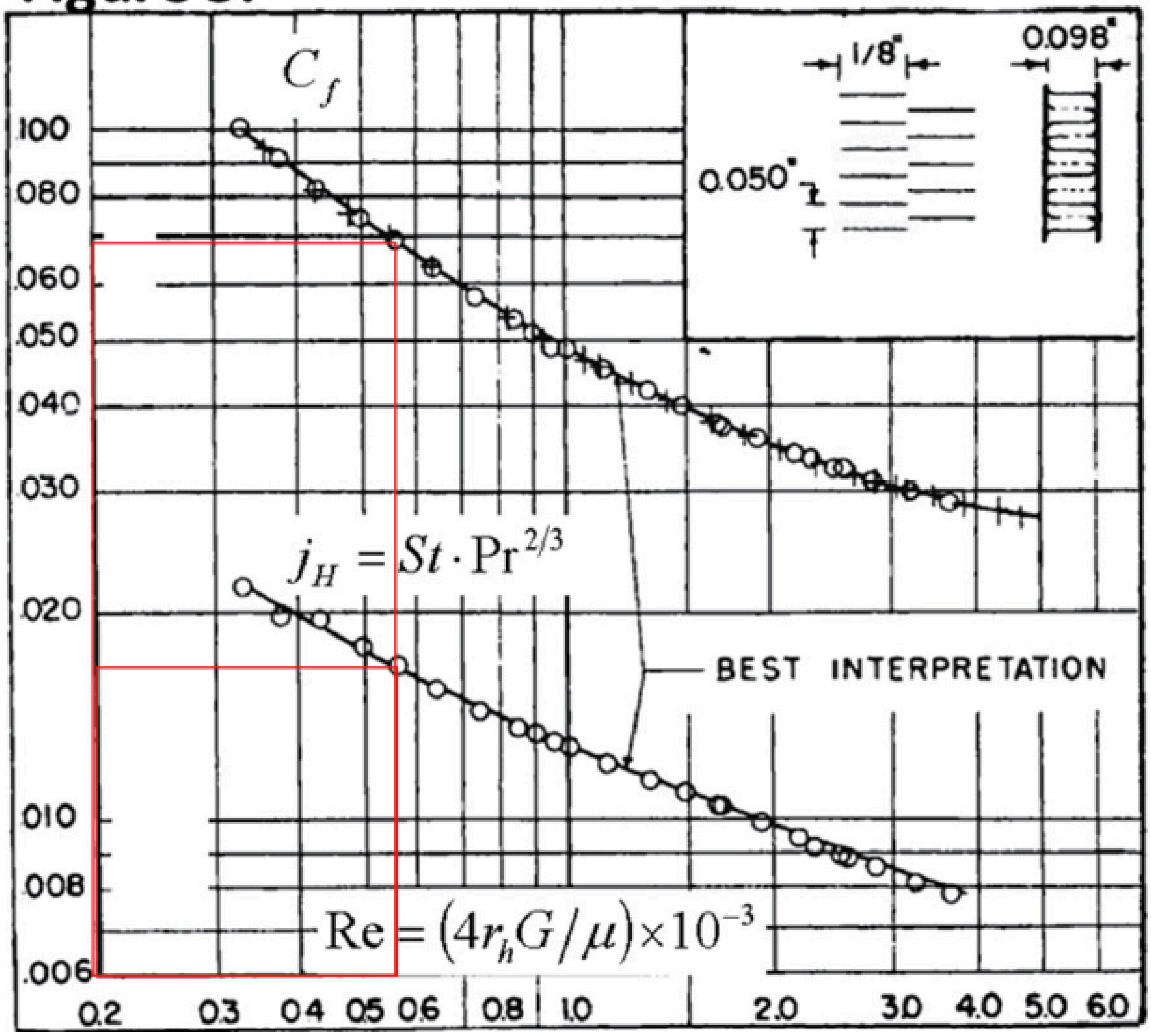
Appendix



Iteration 1:

Gas: Re = 559.41

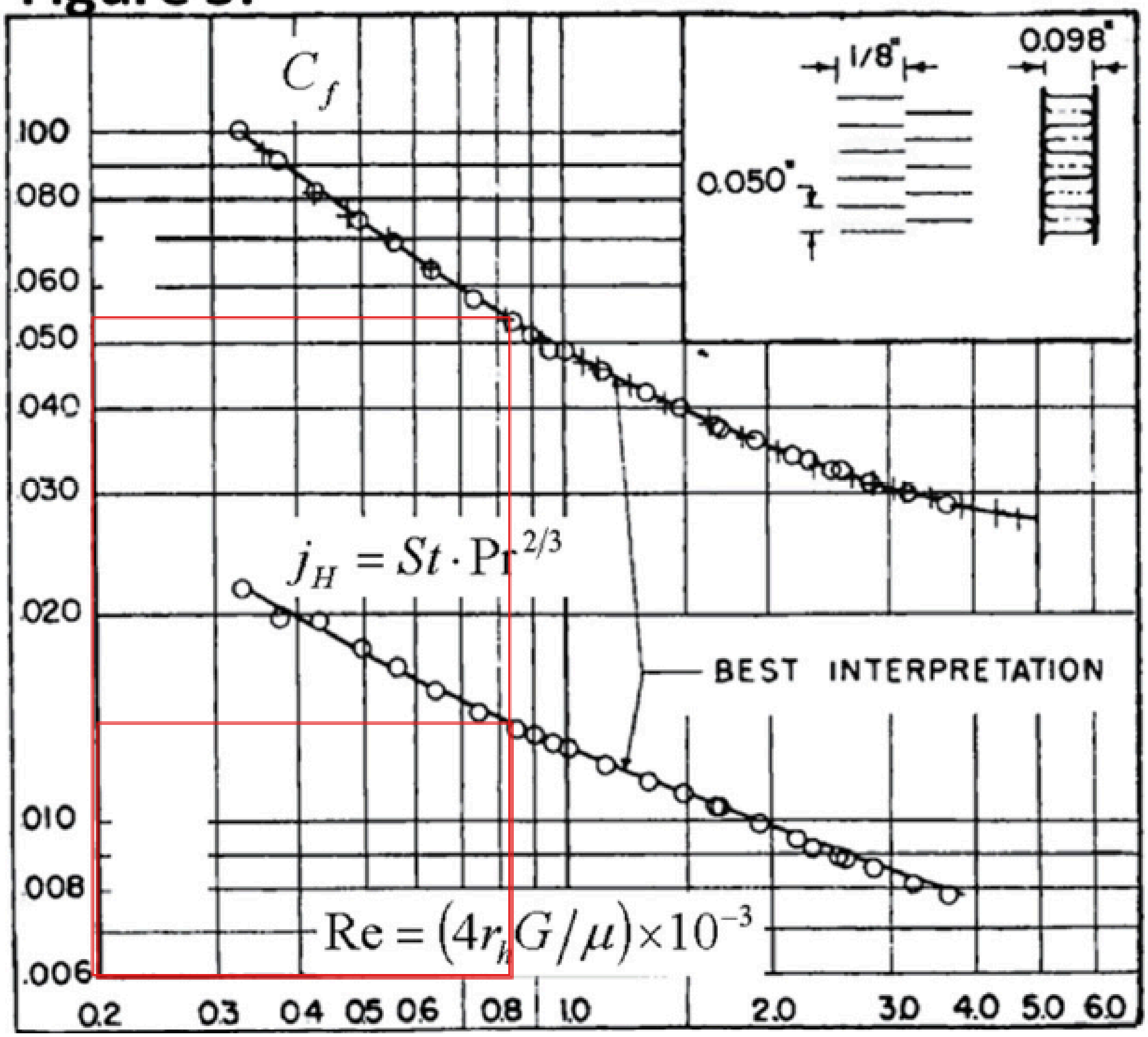
Figure 3:



Iteration 1:

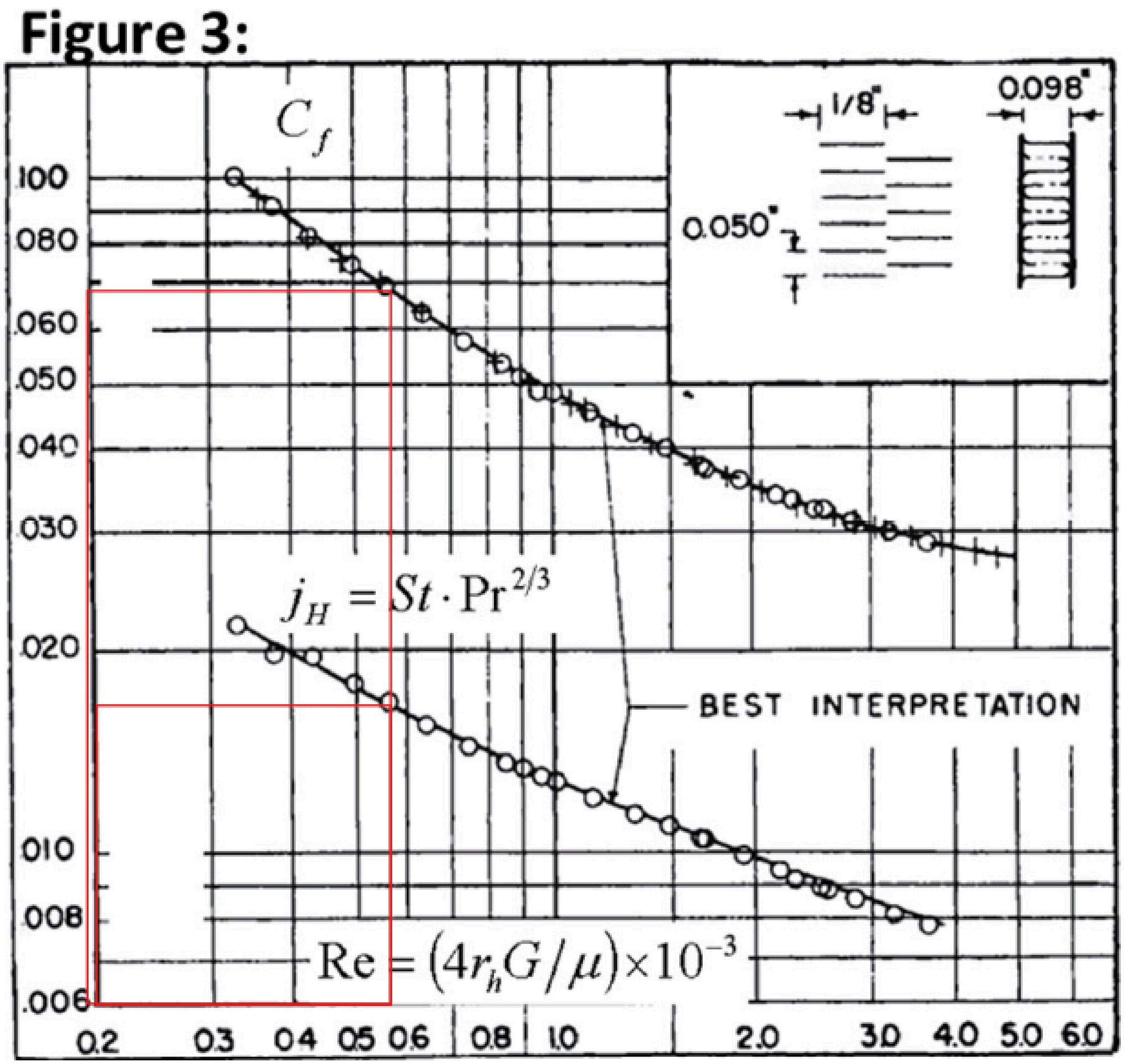
Air: Re = 838.82

Figure 3:



Iteration 2:

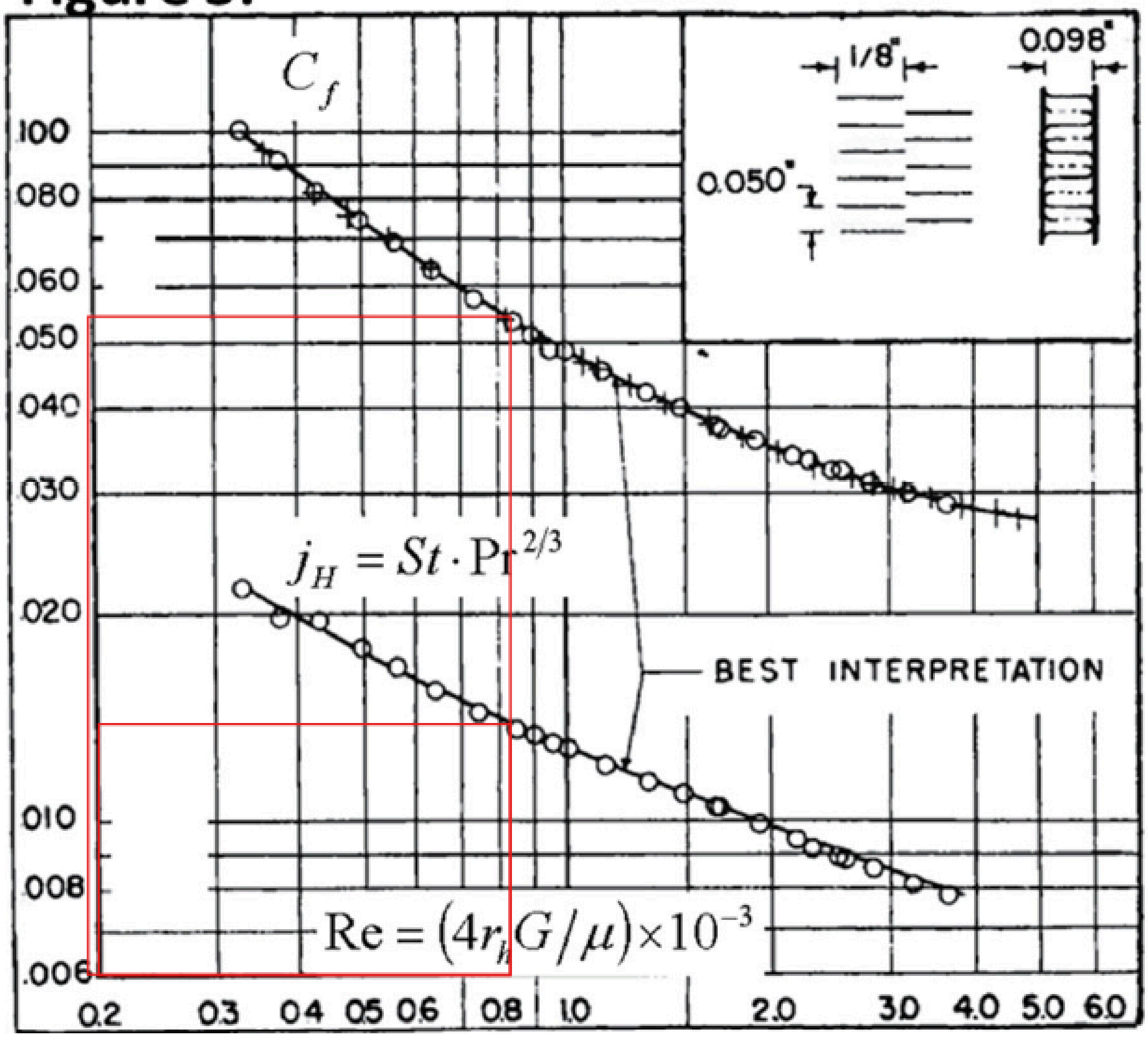
Gas: Re = 568



Iteration 2:

Air: Re = 825

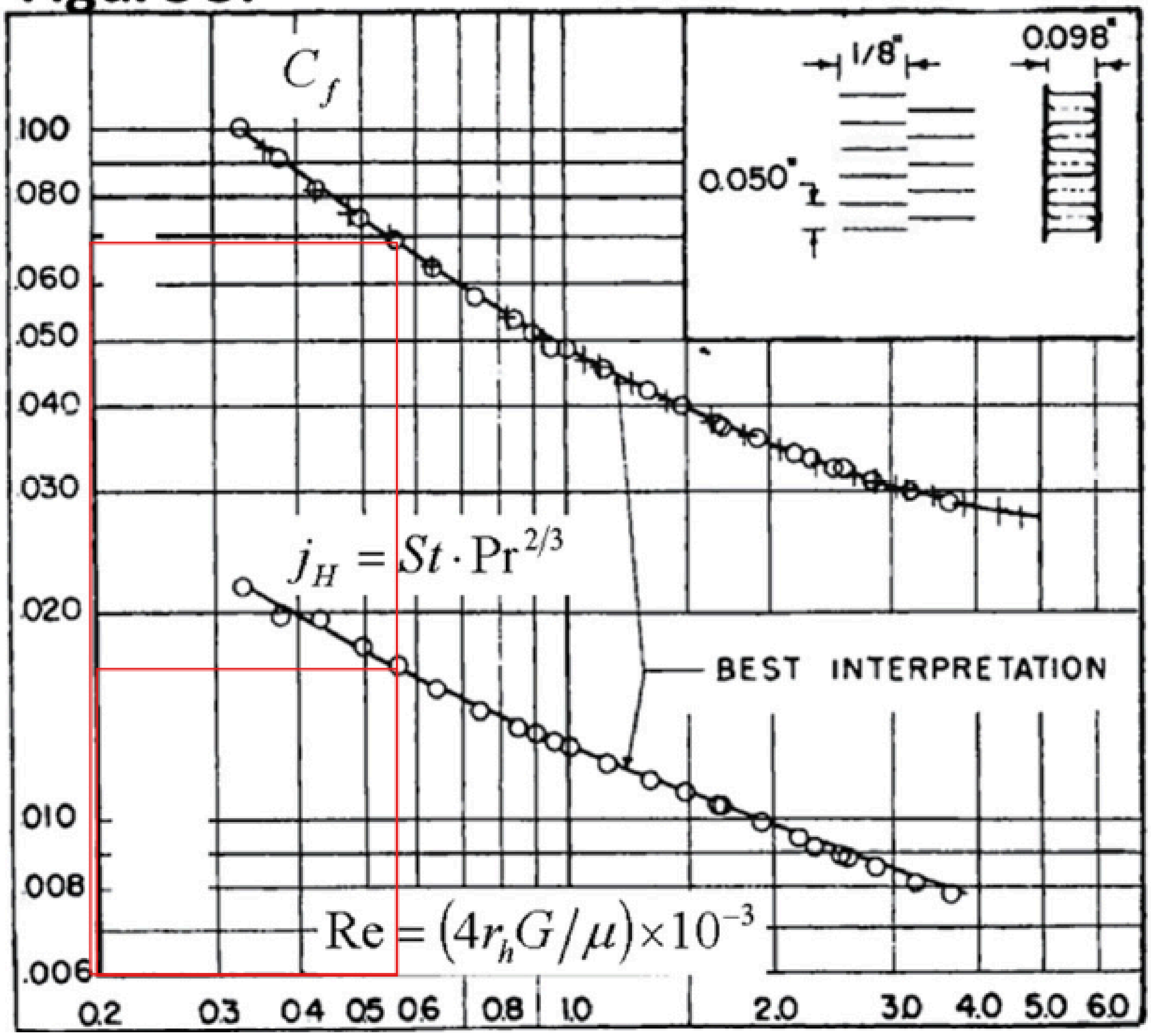
Figure 3:



Iteration 3:

Gas: Re = 560

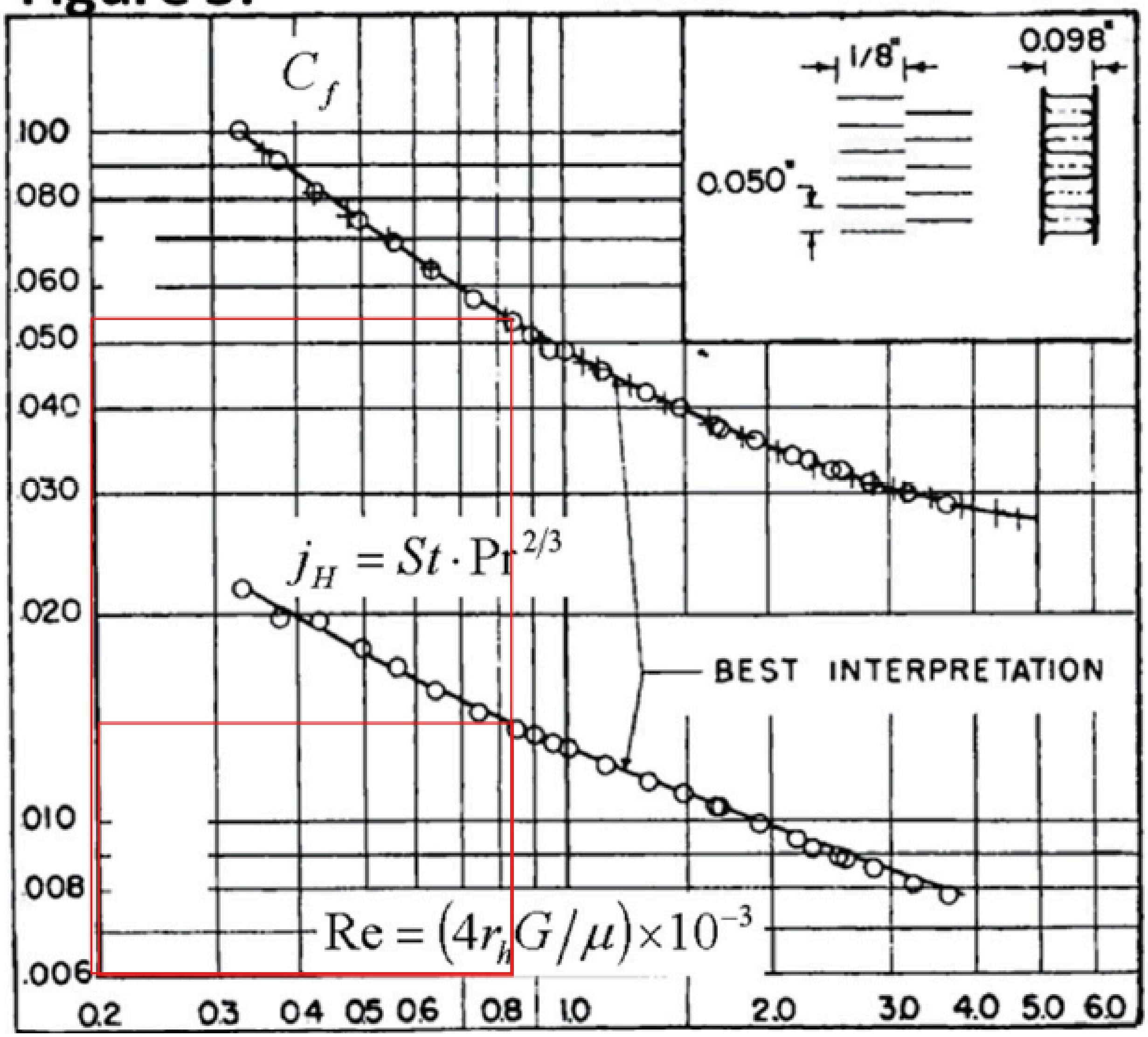
Figure 3:



Iteration 3:

Air: Re = 824.36

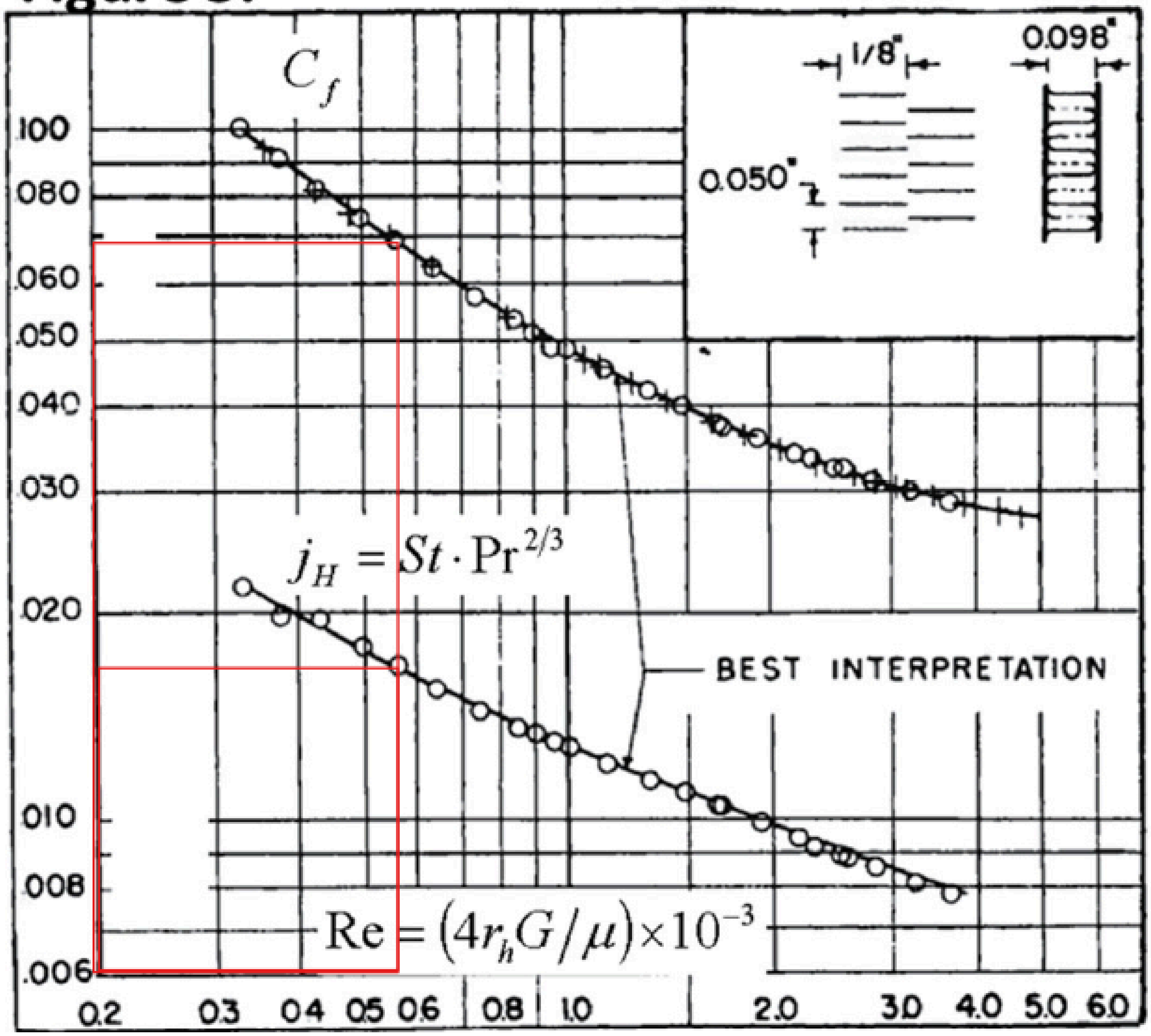
Figure 3:



Iteration 4:

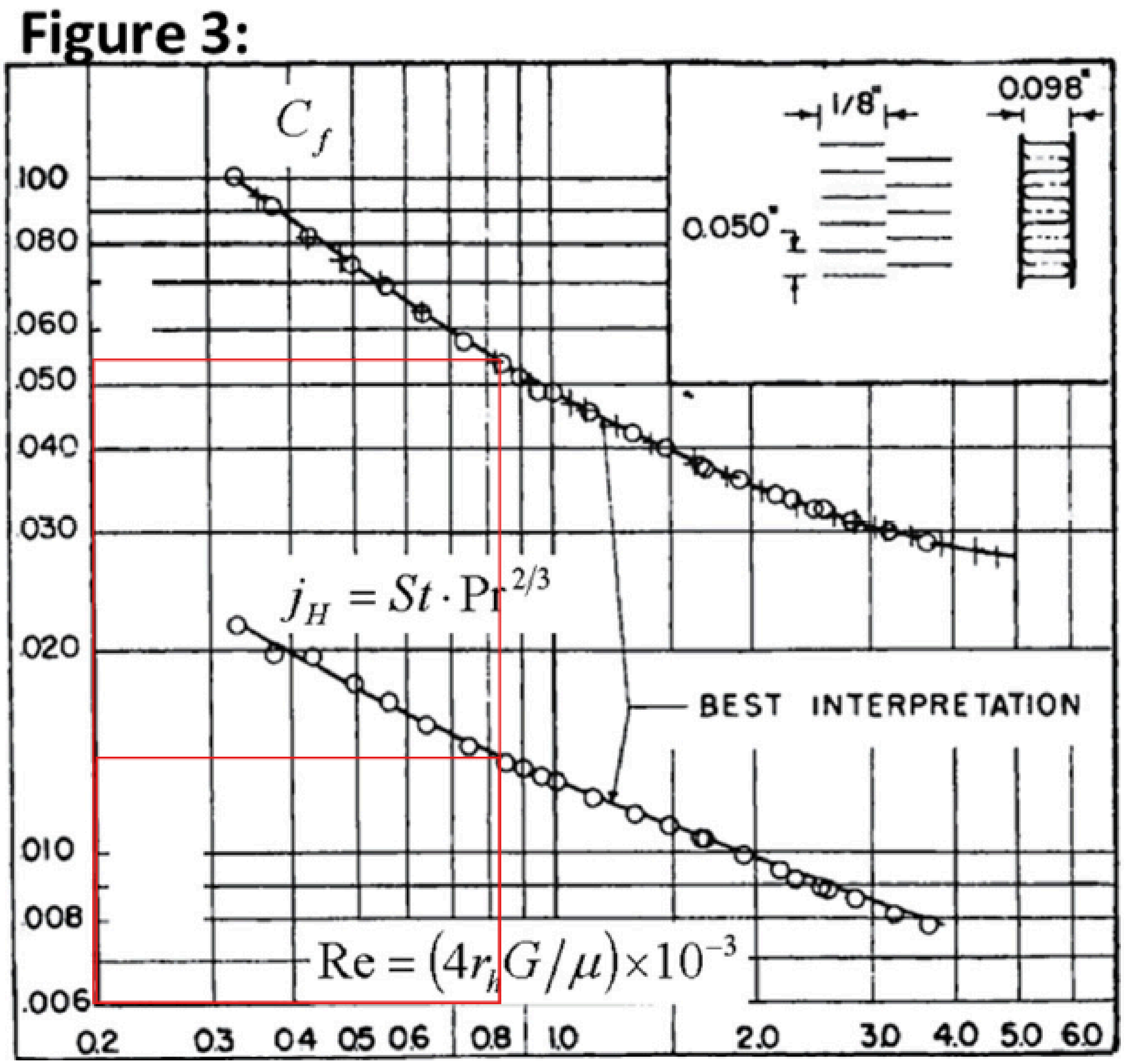
Gas: Re = 560.35

Figure 3:



Iteration 4:

Air: Re = 824.36



MATLAB Code for iterative process:

```
%% constants
sigma = 0.363;
kc = 0.36;
ke = 0.42;
rh = 0.385*10^-3;
l = 0.001143;
Dh = 0.00154;
Cmin = 1853.44;
Cmax = 2174.5;
NTU = 7.79;
%% hot gas
G_h = 14.028;
rho_in_h = 0.4751;
rho_out_h = 0.9504;
specificmass_h = 0.5*((1/rho_in_h)+(1/rho_out_h));
Cf_h = 0.07;
L_h = 0.311;
mew_h = 0.0000392;
cp_h = 1116.5;
Pr_h = 0.718;
A = 95.73;
Ao_h = 0.1183;
Afr_h = 0.326;
%% cold air
G_c = 17.2118;
rho_in_c = 1.4728;
rho_out_c = 0.7160;
specificmass_c = 0.5*((1/rho_in_c)+(1/rho_out_c));
Cf_c = 0.056;
L_c = 0.317;
mew_c = 0.0000346;
cp_c = 1080.53;
Pr_c = 0.698;
A = 95.73;
Ao_c = 0.1162;
Afr_c = 0.32;
%% initial pressure drops
delta_P_h0 = (G_h^2 / (2*rho_in_h))*(1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke)))
delta_P_c0 = (G_c^2 / (2*rho_in_c))*(1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke)))
```

```
%% iteration 1
delta_P_spec_h = 9050;
G_h = ((2*rho_in_h*delta_P_spec_h)/ (1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke))))^0.5;
Re = G_h*Dh/mew_h;
%for Re = 559.41
jH = 0.017;
Cf_h = 0.068;
h = jH*G_h*cp_h*Pr_h^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l)
eff_o = 1-((0.785)*(1-eff_f));
eff_o_h = eff_o;
h_h = h;
delta_P_spec_c = 8790;
G_c = ((2*rho_in_c*delta_P_spec_c)/ (1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke))))^0.5;
Re = G_c*Dh/mew_c;
%for Re = 838.82
jH = 0.0145;
Cf_c = 0.054;
h = jH*G_c*cp_c*Pr_c^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l);
eff_o = 1-((0.785)*(1-eff_f));
eff_o_c = eff_o;
h_c = h;
% U calculation
L_3 = Afr_h/L_c;
Np = floor((L_3-0.00249-(2*0.0005))/(0.00249+0.00249+(2*0.0005)));
Aw = L_c*L_h*((2*Np)+2);
U = 1/((1/(eff_o_c*h_c))+(1/(eff_o_h*h_h))+((0.0005*A)/(18*Aw)));
U = round(U,2);
%core dimensions hot
A = NTU*Cmin/U;
Ao_h = 1.66/G_h;
Afr_h = Ao_h/sigma;
L = (0.00154*A)/(4*Ao_h);
L_h = L;
%core dimensions cold
A = NTU*Cmin/U;
Ao_c = 2/G_c;
Afr_c = Ao_c/sigma;
L = (0.00154*A)/(4*Ao_c);
```

```
L_c = L;
%Pressure drops - iteration 1
delta_P_h1 = (G_h^2 / (2*rho_in_h))*(1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke)))
delta_P_c1 = (G_c^2 / (2*rho_in_c))*(1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke)))
%% iteration 2
delta_P_spec_h = 9050;
G_h = ((2*rho_in_h*delta_P_spec_h)/ (1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke))))^0.5;
Re = G_h*Dh/mew_h;
%for Re = 568
jH = 0.0173;
Cf = 0.069;
h = jH*G_h*cp_h*Pr_h^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l)
eff_o = 1-((0.785)*(1-eff_f));
eff_o_h = eff_o;
h_h = h;
delta_P_spec_c = 8790;
G_c = ((2*rho_in_c*delta_P_spec_c)/ (1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke))))^0.5;
Re = G_c*Dh/mew_c;
%for Re = 825
jH = 0.0147;
Cf = 0.055;
h = jH*G_c*cp_c*Pr_c^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l);
eff_o = 1-((0.785)*(1-eff_f));
eff_o_c = eff_o;
h_c = h;
% U calculation
L_3 = Afr_h/L_c;
Np = floor((L_3-0.00249-(2*0.0005))/(0.00249+0.00249+(2*0.0005)));
Aw = L_c*L_h*((2*Np)+2);
U = 1/((1/(eff_o_c*h_c))+(1/(eff_o_h*h_h))+((0.0005*A)/(18*Aw)));
U = round(U,2);
%core dimensions hot
A = NTU*Cmin/U;
Ao_h = 1.66/G_h;
Afr_h = Ao_h/sigma;
L = (0.00154*A)/(4*Ao_h);
```

```
L_h = L;
%core dimensions cold
A = NTU*Cmin/U;
Ao_c = 2/G_c;
Afr_c = Ao_c/sigma;
L = (0.00154*A)/(4*Ao_c);
L_c = L;
%Pressure drops - iteration 1
delta_P_h2 = (G_h^2 / (2*rho_in_h))*(1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke)))
delta_P_c2 = (G_c^2 / (2*rho_in_c))*(1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke)))
%% iteration 3
delta_P_spec_h = 9050;
G_h = ((2*rho_in_h*delta_P_spec_h)/ (1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke))))^0.5;
Re = G_h*Dh/mew_h;
%for Re = 560
jH = 0.0175;
Cf = 0.07;
h = jH*G_h*cp_h*Pr_h^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l)
eff_o = 1-((0.785)*(1-eff_f));
eff_o_h = eff_o;
h_h = h;
delta_P_spec_c = 8790;
G_c = ((2*rho_in_c*delta_P_spec_c)/ (1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke))))^0.5;
Re = G_c*Dh/mew_c;
%for Re = 824.36
jH = 0.0147;
Cf = 0.055;
h = jH*G_c*cp_c*Pr_c^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l);
eff_o = 1-((0.785)*(1-eff_f));
eff_o_c = eff_o;
h_c = h;
% U calculation
L_3 = Afr_h/L_c;
Np = floor((L_3-0.00249-(2*0.0005))/(0.00249+0.00249+(2*0.0005)));
Aw = L_c*L_h*((2*Np)+2);
U = 1/((1/(eff_o_c*h_c))+(1/(eff_o_h*h_h))+((0.0005*A)/(18*Aw)));
```

```
U = round(U,2);
%core dimensions hot
A = NTU*Cmin/U;
Ao_h = 1.66/G_h;
Afr_h = Ao_h/sigma;
L = (0.00154*A)/(4*Ao_h);
L_h = L;
%core dimensions cold
A = NTU*Cmin/U;
Ao_c = 2/G_c;
Afr_c = Ao_c/sigma;
L = (0.00154*A)/(4*Ao_c);
L_c = L;
%Pressure drops - iteration 1
delta_P_h3 = (G_h^2 / (2*rho_in_h))*(1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke)))
delta_P_c3 = (G_c^2 / (2*rho_in_c))*(1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke)))
%% iteration 4
delta_P_spec_h = 9050;
G_h = ((2*rho_in_h*delta_P_spec_h)/ (1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke))))^0.5;
Re = G_h*Dh/mew_h;
%for Re = 560.35
jH = 0.0175;
Cf = 0.07;
h = jH*G_h*cp_h*Pr_h^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l)
eff_o = 1-((0.785)*(1-eff_f));
eff_o_h = eff_o;
h_h = h;
delta_P_spec_c = 8790;
G_c = ((2*rho_in_c*delta_P_spec_c)/ (1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke))))^0.5;
Re = G_c*Dh/mew_c;
%for Re = 824.36
jH = 0.0147;
Cf = 0.055;
h = jH*G_c*cp_c*Pr_c^(-2/3);
m = ((h*0.006564)/(18*0.32436*10^-6))^0.5;
eff_f = tanh(m*l)/(m*l);
eff_o = 1-((0.785)*(1-eff_f));
eff_o_c = eff_o;
```

```
h_c = h;
% U calculation
L_3 = Afr_h/L_c;
Np = floor((L_3-0.00249-(2*0.0005))/(0.00249+0.00249+(2*0.0005)));
Aw = L_c*L_h*((2*Np)+2);
U = 1/((1/(eff_o_c*h_c))+(1/(eff_o_h*h_h))+((0.0005*A)/(18*Aw)));
U = round(U,2);
%core dimensions hot
A = NTU*Cmin/U;
Ao_h = 1.66/G_h;
Afr_h = Ao_h/sigma;
L = (0.00154*A)/(4*Ao_h);
L_h = L;
%core dimensions cold
A = NTU*Cmin/U;
Ao_c = 2/G_c;
Afr_c = Ao_c/sigma;
L = (0.00154*A)/(4*Ao_c);
L_c = L;
%Pressure drops - iteration 1
delta_P_h4 = (G_h^2 / (2*rho_in_h))*(1 - sigma^2 +kc+ (2*((rho_in_h/rho_out_h) -1
))+(Cf_h*L_h*rho_in_h*specificmass_h/rh)-((rho_in_h/rho_out_h)*(1-sigma^2 -ke)))
delta_P_c4 = (G_c^2 / (2*rho_in_c))*(1 - sigma^2 +kc+ (2*((rho_in_c/rho_out_c) -1
))+(Cf_c*L_c*rho_in_c*specificmass_c/rh)-((rho_in_c/rho_out_c)*(1-sigma^2 -ke)))
clc
%Pressure drop after 4th iteration has an error less than 1 percent from
%give values of pressure drop
delta_P_h1
delta_P_c1
delta_P_h2
delta_P_c2
delta_P_h3
delta_P_c3
delta_P_gas_final = delta_P_h4
delta_P_air_final = delta_P_c4
Error_percent_gas = abs(delta_P_gas_final-delta_P_spec_h)/delta_P_spec_h * 100
Error_percent_air = abs(delta_P_air_final-delta_P_spec_c)/delta_P_spec_c * 100
%Core dimensions of Heat Exchanger
Lg = L_h
La = L_c
L = L_3
A
```